

Differentiable solvers for time series segmentation

Time series segmentation — change-point detection, piecewise approximation, regime discovery — rests on discrete optimization that is non-differentiable, blocking its use as a layer in trainable pipelines. Smoothing a fixed-grid dynamic program is well understood; the real obstacle is *model selection*, since the optimizer also chooses how many segments to use. This internship asks whether that segment-count decision can itself be made differentiable, end to end.

§1. Organization

- **Start date:** As soon as possible (flexible).
- **Localization:** IRISA, Rennes.
- **Supervision:** Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).

§2. Context

Time series segmentation is a family of related problems that partition a signal into homogeneous or meaningful pieces. It includes change-point detection / optimal partitioning (locating shifts in the statistical properties of a signal, solved e.g. by dynamic-programming methods), piecewise approximation (representing a signal as a sequence of constant or linear segments, used for compression, indexing, and interpretable summaries), and regime- or motif-based segmentation (grouping the signal into recurring states) (Truong et al., 2020). All of these reduce to discrete optimization over candidate breakpoints or partitions, typically solved by dynamic programming or greedy heuristics — and all are *non-differentiable*.

Non-differentiability prevents these solvers from being used as layers inside end-to-end trainable pipelines: one cannot learn a representation or cost function whose induced segmentation is better for a downstream task, nor jointly train a forecasting or anomaly-detection model that exploits inferred segment structure. Generic toolboxes for differentiating combinatorial solvers — continuous relaxation (Mensch & Blondel, 2018) and stochastic perturbation (Berthet et al., 2020) — exist, and a differentiable change-point detector has been proposed for one specific paradigm (Koley et al., 2023).

§3. Scientific challenge

The fixed-trellis case is the easy one. When the segmentation grid is fixed — as in dynamic time warping — smoothing the dynamic program is well understood. Segmentation’s genuine difficulty is *model selection*: the number of segments is not given but is itself chosen by the optimizer, typically by trading a data-fit term against a penalty on the number of breakpoints. This makes the solution an integer-valued, piecewise-constant function of both the signal and the penalty, so the segment count jumps discontinuously and standard smoothing of the inner DP does not produce a gradient through the *selection* of how many segments to use. Recent work learns this penalty with a separate predictor (Truong & Runge, 2025), but treats the partitioning solver itself as a fixed, non-differentiable oracle. The open question is therefore: **can the model-selection step that controls segment count be made differentiable, so that a representation, cost function, and penalty are learned jointly and end-to-end?** A secondary question is whether relaxation or perturbation handles this discontinuity more stably, and whether the answer differs across segmentation paradigms.

§4. Goals

Concretely, the candidate will:

1. formalize two or three representative segmentation paradigms as penalized DP / combinatorial problems (optimal partitioning / change-point detection; piecewise-constant or -linear approximation; optionally a regime-based variant), making the penalty / segment-count choice explicit;
2. derive relaxation- and/or perturbation-based gradients that flow through both the segment costs *and* the model-selection step, with respect to the signal, a learned cost function, and the penalty;
3. benchmark accuracy, gradient quality, and speed against classical and learned discrete solvers using a unified segmentation toolkit (Chavelli et al., 2026) on its built-in datasets and metrics;
4. demonstrate an end-to-end use case — jointly learning a feature extractor and penalty with a differentiable segmentation layer (e.g. on regime-switching series or an anomaly-detection benchmark);
5. (stretch goal) compare paradigms on how stably their model-selection step can be relaxed or perturbed.

Candidates should possess:

- good Python coding skills,
- knowledge of dynamic programming and time-series fundamentals,
- broader familiarity with time-series representations is a plus,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a practical and theoretical understanding of differentiable combinatorial optimization,
- knowledge of how segmentation paradigms map to dynamic programs,
- experience designing benchmarks and end-to-end training pipelines for time series.

§5. Bibliography

- Berthet, Q., Blondel, M., Teboul, O., Cuturi, M., Vert, J.-P., & Bach, F. (2020). *Learning with Differentiable Perturbed Optimizers*. NeurIPS.
- Chavelli, F., Ermshaus, A., Yang, F., Schäfer, P., Paparrizos, J., et al. (2026). *tsseg: An Interactive Toolkit for Time Series Segmentation*. ECML-PKDD.
- Koley, P., De, A., Ganguly, N., et al. (2023). *Differentiable Change-point Detection with Temporal Point Processes*. AISTATS.
- Mensch, A., & Blondel, M. (2018). *Differentiable Dynamic Programming for Structured Prediction and Attention*. ICML.
- Truong, C., Oudre, L., & Vayatis, N. (2020). *Selective Review of Offline Change Point Detection Methods*. Signal Processing.
- Truong, C., & Runge, V. (2025). *Learning Penalty for Optimal Partitioning via Automatic Feature Extraction*. arXiv:2505.07413.